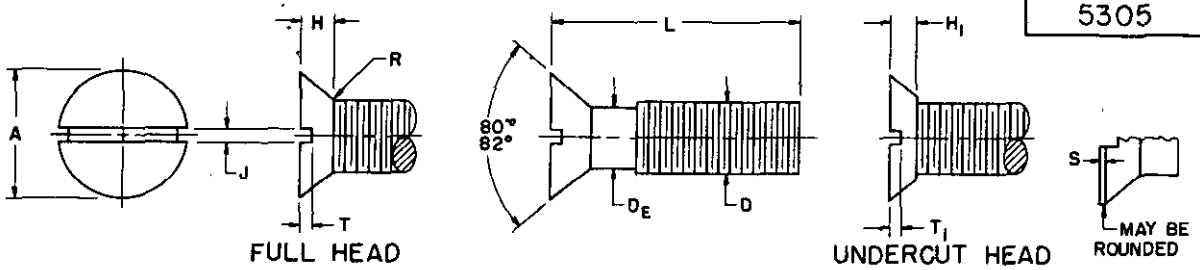
FED. SUP CLASS
5305



			#1 (.0730)	#2 (.0860)	#4 (.1120)	#6 (.1380)	#8 (.1640)	#10 (.1900)	1/4 (.2500)	5/16 (.3125)
Nominal Size D										
Threads Per Inch			64NC	56NC	40NC	32NC	32NC	24NC	20UNC	18UNC
Body Diameter D_E		Max	.0730	.0860	.1120	.1380	.1640	.1900	.2500	.3125
		Min	.0603	.0717	.0925	.1141	.1399	.1586	.2127	.2712
Head Diameter A		Max Sharp	.146	.172	.225	.279	.332	.385	.507	.635
		Min Sharp	.130	.156	.207	.257	.308	.359	.477	.600
		Abs Min	.126	.150	.200	.249	.300	.348	.462	.581
Slot Width J		Max	.026	.031	.039	.048	.054	.060	.075	.084
		Min	.019	.023	.031	.039	.045	.050	.064	.072
Flat		S Max	.003	.003	.004	.005	.006	.007	.009	.011
Radius		R Max	.031	.036	.050	.063	.063	.083	.100	.111
Full Head	Head Height H	Max	.043	.051	.067	.083	.100	.116	.153	.191
		Min	.033	.040	.055	.069	.084	.098	.131	.165
	Slot Depth T	Max	.019	.023	.030	.038	.045	.053	.070	.088
		Min	.012	.015	.020	.024	.029	.034	.046	.058
Undercut Head	Head Height H_1	Max	.032	.037	.048	.060	.071	.083	.108	.135
		Min	.022	.026	.036	.046	.055	.065	.086	.109
	Slot Depth T_1	Max	.016	.019	.025	.032	.037	.044	.058	.073
		Min	.012	.015	.020	.024	.029	.034	.046	.058

Length L	Tolerance	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.	Dash No.
1/8		1	8	18	31	46			
3/16		2	9	19	32	47	65		
1/4		3	10	20	33	48	66		
5/16		4	11	21	34	49	67	83	
3/8	+0	5	12	22	35	50	68	84	99
7/16	-1/32	6	13	23	36	51	69	85	100
1/2		7	14	24	37	52	70	86	101
5/8			15	25	38	53	71	87	102
3/4			16	26	39	54	72	88	103
7/8			17	27	40	55	73	89	104
1				28	41	56	74	90	105
1-1/4	+0			29	42	57	75	91	106
1-1/2	-1/16			30	43	58	76	92	107
1-3/4					44	59	77	93	108
2					45	60	78	94	109
2-1/4	+0					61	79	95	110
2-1/2	-3/32			Ⓐ CANCELLED		62	80	96	111
2-3/4				SEE SHEET 2		63	81	97	112
3						64	82	98	113

Threads shall extend to within 2 threads of the bearing surface of the head, or closer if practicable.

Minimum complete thread length of 1-3/4.

MATERIAL: Carbon steel, Specification QQ-W-409 or QQ-S-633, except Bessemer compositions; 55,000 PSI minimum ultimate tensile strength.
FINISH: Cadmium plated, Specification QQ-P-416, type II, class C.
THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services Handbook H-28.
NOTES: (1) Referenced documents of issue in effect on date of invitation for bids shall apply.
(2) In case of conflict with any referenced document, this standard will govern.
(3) The MS part number consists of the MS sheet number, plus the dash number. Example: MS 35239-1.
(4) Sizes above dotted line have undercut head.
(5) All dimensions in inches.

Ⓐ For changes see sheets 1 and 2.

APPROVED FEB 9, 1954 REVISED Ⓐ 5 APRIL 1962

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. ORD
Other Cust - SHIPS

TITLE
SCREW, MACHINE, FLAT COUNTERSUNK HEAD, CARBON STEEL, CADMIUM PLATED, NC-2A AND UNC-2A.

MILITARY STANDARD
MS 35239

PROCUREMENT SPECIFICATION
FF-S-92

SUPERSEDES:

SHEET 1 OF 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:12Z
Check the source to verify that this is the current version before use.

DD FORM 1 SEP 57 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

FED. SUP CLASS 5305

NOMINAL SIZE D			3/8 (.3750)	7/16 (.4375)	1/2 (.5000)	9/16 (.5625)	5/8 (.6250)	3/4 (.7500)		
THREADS PER INCH			16UNC	14UNC	13UNC	12UNC	11UNC	10UNC		
BODY DIAMETER D_E		MAX	.3750	.4375	.5000	.5625	.6250	.7500		
		MIN	.3287	.3850	.4435	.5016	.5589	.6773		
HEAD DIAMETER A		MAX SHARP	.762	.812	.875	1.000	1.125	1.375		
		MIN SHARP	.722	.771	.831	.950	1.069	1.306		
		ABS MIN	.700	.743	.802	.919	1.035	1.267		
SLOT WIDTH J		MAX	.094	.094	.106	.118	.133	.149		
		MIN	.081	.081	.091	.102	.116	.131		
FLAT S		MAX	.013	.016	.018	.020	.023	.027		
RADIUS R		MAX	.125	.143	.154	.167	.182	.200		
FULL HEAD	HEAD HEIGHT H	MAX	.230	.223	.223	.260	.298	.372		
		MIN	.200	.190	.186	.220	.253	.319		
	SLOT DEPTH T	MAX	.106	.103	.103	.120	.137	.171		
		MIN	.070	.066	.065	.077	.088	.111		
UNDERCUT HEAD	HEAD HEIGHT H_1	MAX	.162	.156	.160	.185	.212	.264		
		MIN	.132	.126	.123	.145	.167	.211		
	SLOT DEPTH T_1	MAX	.088	.085	.085	.100	.114	.142		
		MIN	.070	.066	.065	.077	.088	.111		

LENGTH L	TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.		
THREADS SHALL EXTEND TO WITHIN 2 THREADS OF THE BEARING SURFACE OF THE HEAD, OR CLOSER IF PRACTICABLE.									
1/8	+0 -1/32	114				Ⓐ CANCELLED SEE NOTE			
3/16		115							
1/4			127						
5/16		116	128	139					
3/8		117	129	140	150		160		
7/16		118	130	141	151		161	170	
1/2		119	131	142	152		162	171	
5/8									
3/4									
7/8									
1									
1-1/4	+0 -1/16	120	132	143	153		163	172	
1-1/2		121	133	144	154		164	173	
1-3/4		122	134	145	155		165	174	
2		123	135	146	156		166	175	
MINIMUM COMPLETE THREAD LENGTH OF 1-3/4, 2-1/4	+0 -3/32								
2-1/2		124	136	147	157		167	176	
2-3/4		125	137	148	158		168	177	
3		126	138	149	159		169	178	

NOTE: FOR ILLUSTRATION, MATERIAL, FINISH, AND OTHER PERTINENT DATA, SEE SHEET 1.

Ⓐ CANCELLED AFTER 7 May 1962 USE MS 35190. INTERCHANGEABILITY RELATIONSHIP-MS 35190 CAN REPLACE THE CANCELLED SCREWS OF THE SAME SIZE. THE CANCELLED MS 35239 CANNOT ALWAYS REPLACE MS 35190 AND SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED. THEN USE ONLY THE SUPERSEDING MS SCREWS OF THE SAME SIZE FOR DESIGN AND REPLACEMENT.

APPROVED FEB 9, 1954 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. ORD Other Cust SHIPS	TITLE SCREW, MACHINE, FLAT COUNTERSUNK HEAD, SLOTTED, CARBON STEEL, CADMIUM PLATED, NC-2A AND UNC-2A.	MILITARY STANDARD MS 35239	
PROCUREMENT SPECIFICATION FF-S-92	SUPERSEDES:	SHEET 2 OF	

DD FORM 1 SEP 57 672-1 (Coordinated) Previous editions of this form are obsolete.

Source: https://assist.dla.mil — Downloaded: 2026-10-23T17:12Z
Check the source to verify that this is the current version before use.